

Curves intersecting at-most once

A construction of a maximal 1-system of curves on the surface of genus 3

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Systems of curves on surfaces

Let Σ_g be the closed orientable surface of genus g .

Definition

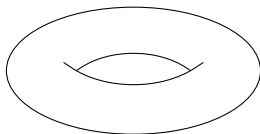
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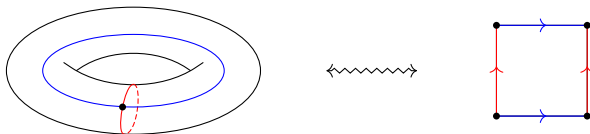


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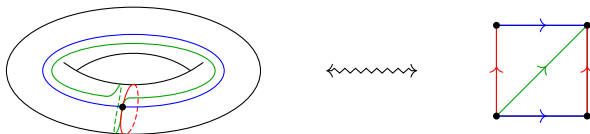


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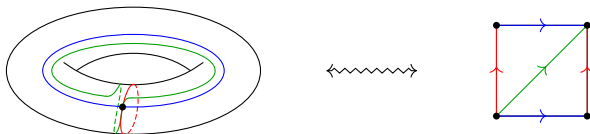


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What is the maximum cardinality N_g of a 1-system on N_g ?

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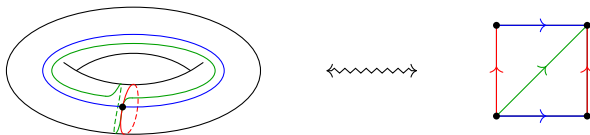


Figure: By some easy algebraic topology, we can see that $N_1 = 3$.

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Contents

① Survey and known results

② The case when $g = 3$

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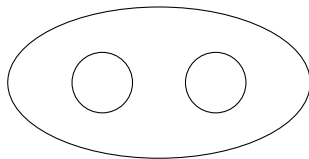
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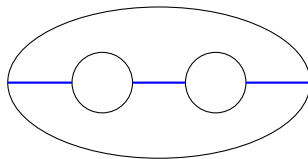
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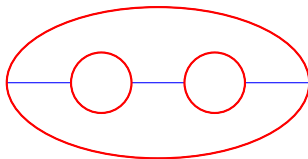
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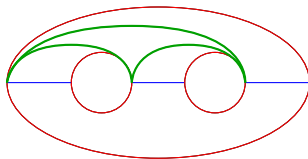
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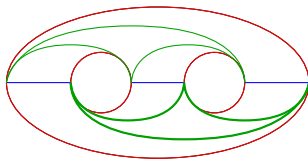
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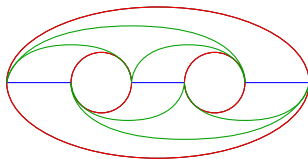
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Theorem (Malestein-Rivin-Theran [MRT14])

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We care about N_g since $\dim(\mathcal{C}^1(\Sigma_g)) = N_g - 1$.

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Finally, Aougab-Gaster [AG25+] very recently showed that $N_g = \Theta(g^2)$.

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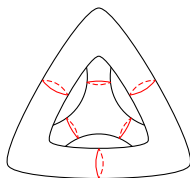


Figure: A sample of the curves in X , where $|X| = 6$

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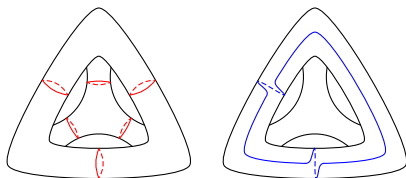


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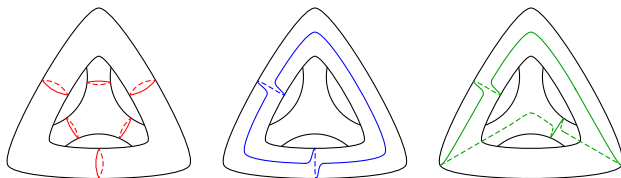


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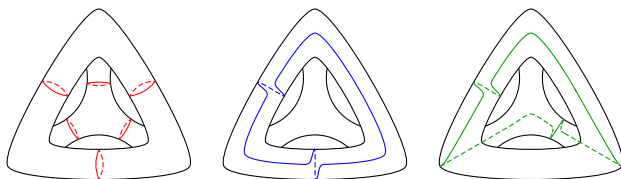


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Every surface Σ_g with $g \geq 2$ can be decomposed into *pairs of pants*, i.e., 3-punctured spheres. A *pants decomposition* of Σ_g is a collection $\mathcal{P}$ of pairs of pants of pants $P \subset \Sigma_g$ such that $\Sigma_g = \bigsqcup \mathcal{P}$.

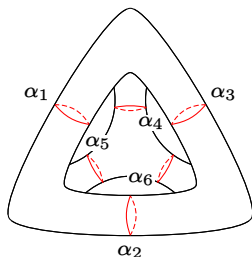


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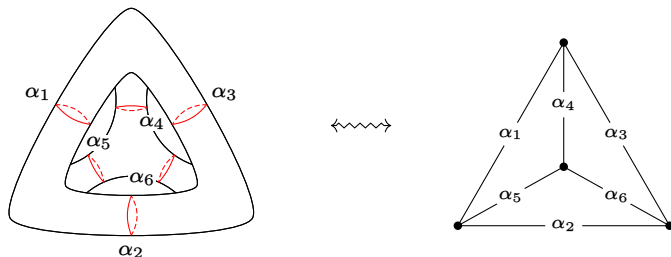


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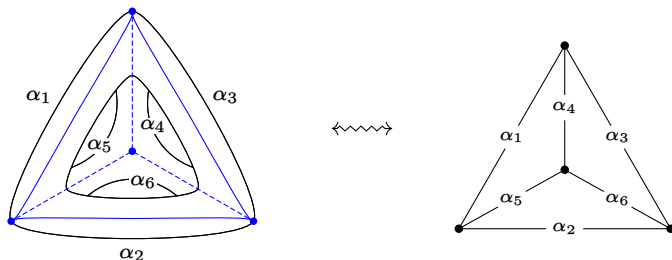
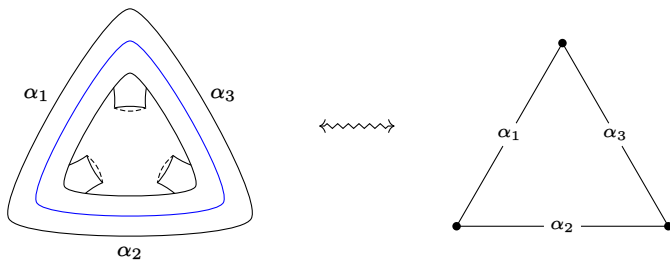


Figure: A pants decomposition $\mathcal{P}$ of Σ_3 and its dual graph $K_4 \hookrightarrow \Sigma_3$.

Curves of type Δ Figure: A reference curve of type Δ .

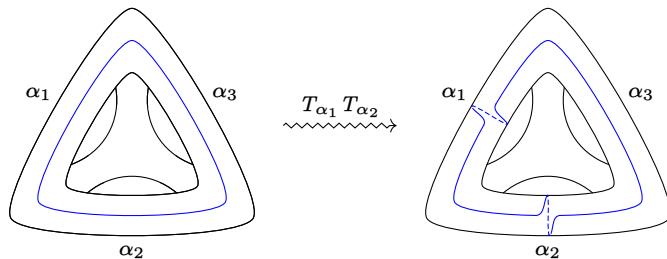
Curves of type Δ 

Figure: A reference curve of type Δ and its multitwist, encoded by $(1, 1, 0)$.

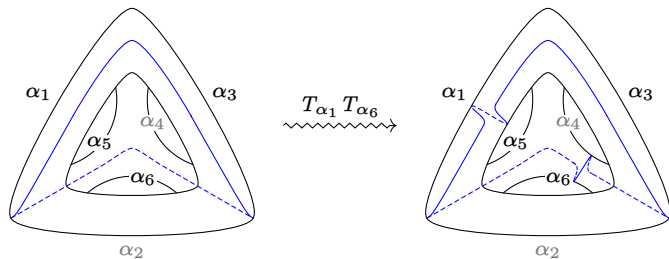
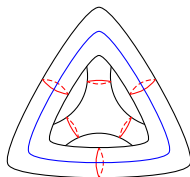
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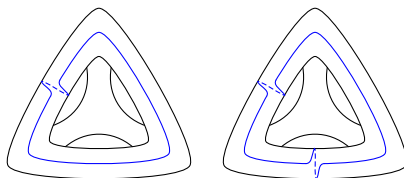
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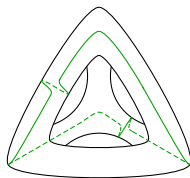
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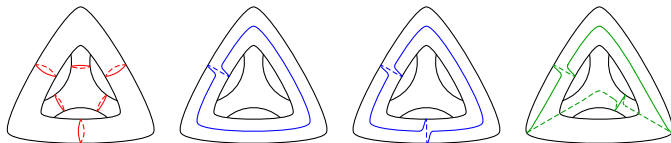
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$$X_0 := \{\alpha_1, \dots, \alpha_6\} \cup \{G\gamma_j : j = 1, 2, 3\},$$

where $G\gamma_j$ is the orbit of γ_j under the group $G := A_4$ acting on $\mathcal{C}_0(\Sigma_3)$ by orientation-preserving symmetries of Σ_3 , is a 1-system of 33 curves on Σ_3 .



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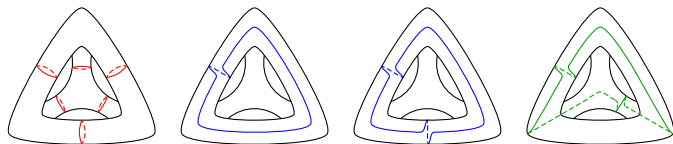
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where $G\gamma_j$ is the orbit of γ_j under the group $G := A_4$ acting on $\mathcal{C}_0(\Sigma_3)$ by orientation-preserving symmetries of Σ_3 , is a 1-system of 33 curves on Σ_3 .

Moreover, X_0 is maximal amongst all 1-systems on Σ_3 .



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Thus the collection $X_\triangle$ of all $6 \times 4 = 24$ curves of type $\triangle$ forms a 1-system.

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4. Prove that the 3 curves $X_\square$ of type $\square$ forms a 1-system.
5. Prove that $X_\triangle \cup X_\square$ is maximal amongst all curves of type $\square$.

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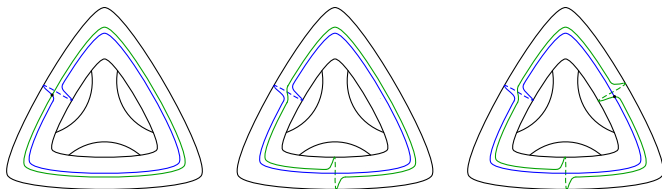
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3. Prove that $X_\triangle$ is maximal amongst all curves of type $\triangle$.
4. Prove that the 3 curves $X_\square$ of type $\square$ forms a 1-system.
5. Prove that $X_\triangle \cup X_\square$ is maximal amongst all curves of type $\square$.

Then $X_0 = \{\alpha_1, \dots, \alpha_6\} \cup X_\triangle \cup X_\square$ is as desired.

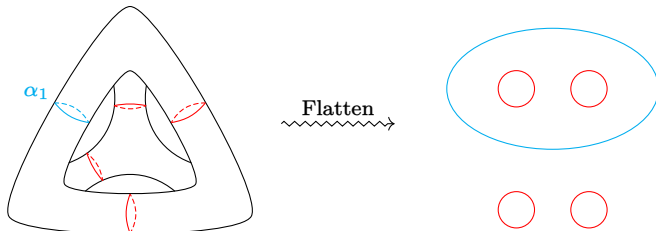
Step 1: Curves of type Δ on one side

We can always preform *Dehn twists* to reduce any configuration of two curves of type Δ on the same face to the following three situations.



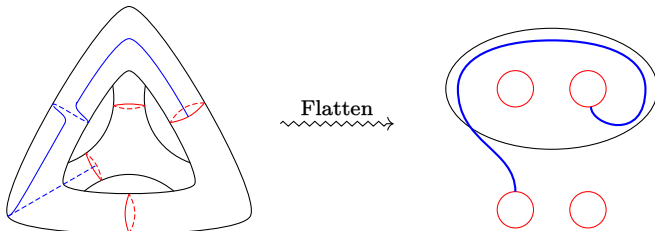
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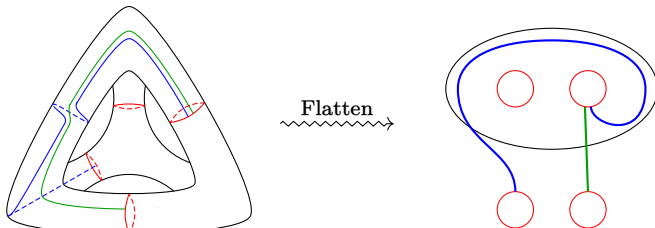
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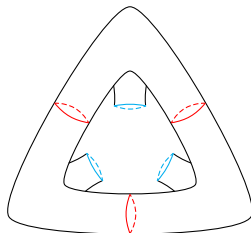
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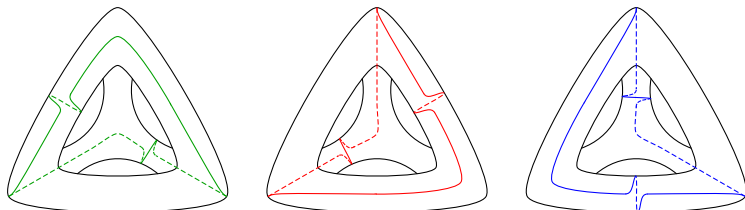
Step 3: Maximality amongst curves of type Δ

If there is a face supporting more than 6 curves of type Δ , then we found a 1-system on the torus with three holes of cardinality at-least $7 + 3 = 10$, but this contradicts [MRT14], stating that the maximum number is $3 \times 3 = 9$.



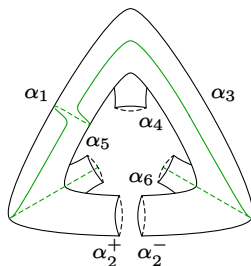
Step 4: Curves of type $\square$ is a 1-system

Under the action of $A_4 \curvearrowright \Sigma_3$, the orbit of $(1, -, 0, -, 0, 1)$ is the following three curves of type $\square$.



Step 5: Maximality amongst curves of type $\square$

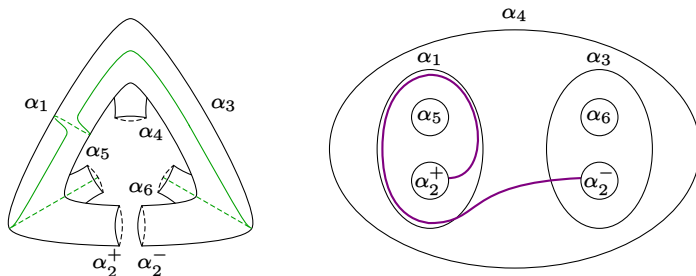
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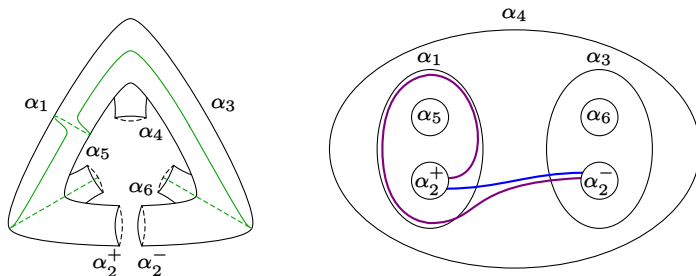
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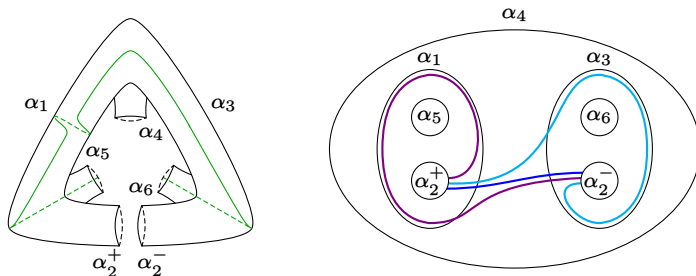
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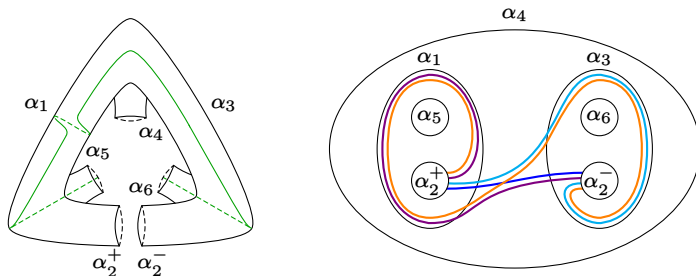
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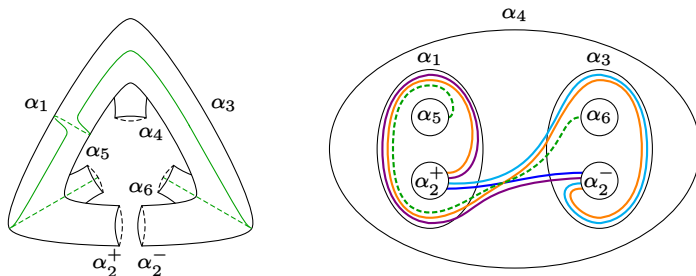
Indeed, no twists can be done in α_1 and α_3 , since, on the 3-cycle induced by $\alpha_1, \dots, \alpha_3$, the existence of the curves $(1, -, 0)$, $(0, -, 0)$, $(0, -, 1)$, and $(1, -, 1)$ of type Δ



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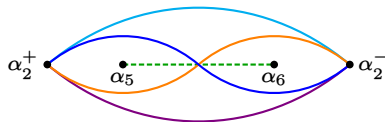
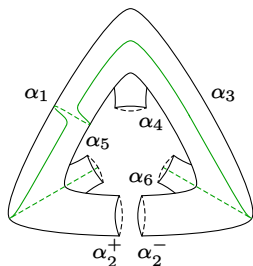
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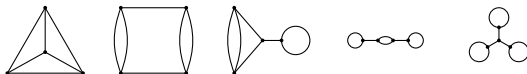
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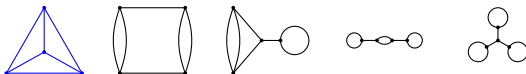
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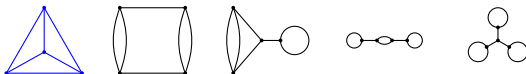
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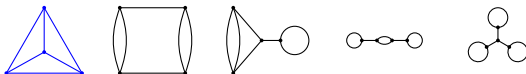
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3. Is the distribution of curves of types $\triangle$ and $\square$ optimal?

Thank you!

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